① $X_1, \dots X_n \sim N(\theta; 4)$ Review class 1 – Solutions

a) 90% confidence interval for the mean

$$[\bar{X} - z_{\alpha/2} \cdot \frac{\sigma}{\sqrt{n}} ; \bar{X} + z_{\alpha/2} \cdot \frac{\sigma}{\sqrt{n}}] =$$

$$= \left[1.45 - \frac{0.51}{4} \cdot z_{0,05} ; 1.45 + \frac{0.51}{4} \cdot z_{0,05}\right] =$$

$$= \left[1.45 - \frac{0.51 \cdot 1.64}{4} ; 1.45 + \frac{0.51 \cdot 1.64}{4}\right] \approx$$

$$[1.45 - 0.2 ; 1.45 + 0.2] = [1.25 ; 1.65]$$

b) With probability 90% true mean belongs to the interval $[1.25; 1.65]$, $2 \notin [1.25; 1.65]$, so H_0 is wrong with probability at least 90%

$\Rightarrow$ we should reject H_0.

c) For the unknown variance,

$$\frac{\bar{X} - \mu}{S/\sqrt{n}} = T \sim t(n-1)$$

$$T = \frac{1.45 - \mu}{0.5/4} = 8(1.45 - \mu) \sim t(15)$$

~~Then $\mathbb{P}(\mu \in [\frac{9}{8}, \frac{5}{4}]) = \mathbb{P}(8(1.45-\mu) \in [8 \cdot 1.45 - 10, 8 \cdot 1.45 - 9]) \approx$~~

~~confidence of $[\frac{9}{8}, \frac{5}{4}]$ $\mathbb{P}(T \in [2, 4])$~~

confidence of $\left[\frac{5}{4}; \frac{13}{8}\right]$ is

$$P\left(\mu \in \left[\frac{5}{4}; \frac{13}{8}\right]\right) = P\left(-8\mu \in [-13; -10]\right) =$$

$$\approx P\left(8(1.45-\mu) \in [-1; 2]\right) = P(T \in [-1, 2])$$

$$= P(T \leq 2) - P(T \leq -1) \approx$$

$$0.97 - (1 - P(T \leq 1)) \approx$$

$$0.97 - 1 + 0.83 = \boxed{0.8}$$

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② $H_0: \mu_0 = 20, \sigma_0 = 4$
$H_1: \mu_1 = 25, \sigma_1 = 5$

$$0.05 = \alpha = P(4 \cdot \bar{x} > \gamma \mid H_0) = P\left(\frac{4\bar{x} - 4\mu_0}{4\sqrt{4}} > \frac{\gamma - 4\mu_0}{4\sqrt{4}} \,\middle|\, H_0\right) =$$

$$= P\left(\underset{\sim N(0,1)}{Z} > \frac{\gamma - 80}{8}\right)$$

$$\frac{\gamma - 80}{8} = z_{0.05} = 1.64 \qquad \gamma = 8 \cdot 1.64 + 80 \approx \underline{93}$$

False acceptance probability $= \beta =$

$$= P(4\bar{x} \leq \gamma \mid H_1) = P\left(\frac{4\bar{x} - 4\mu_1}{\sqrt{4} \cdot 5} \leq \frac{\gamma - 4\mu_1}{5 \cdot 2} \,\middle|\, H_1\right)$$

$$= P\left(\underset{\sim N(0,1)}{Z} \leq \frac{\gamma - 100}{10}\right) \approx \Phi\left(\frac{93 - 100}{10}\right) = \Phi(-0.7) \approx \underline{0.242}$$

③ $X_1 \dots X_{10} \sim Poi(\mu)$

$H_0: \mu = 0.5$ vs $H_1: \mu > 0.5$

By Neymen-Pearson lemma, critical region C is ~~uniformly~~ best for testing $H_0: \mu = 0.5$ vs $\hat{H}_1: \mu = \mu_1 > 0.5$, if

$$\frac{L(0.5)}{L(\mu)} \le k \text{ for any } x \in C$$

$$\frac{L(0.5)}{L(\mu)} = \frac{0.5^{\Sigma x_i} \cdot e^{-n/2}}{x_1! \dots x_n!} \cdot \frac{x_1! \dots x_n!}{\mu_1^{\Sigma x_i} e^{-\mu_1 n}} =$$

$$= \left(\frac{1}{2\mu_1}\right)^{n\bar{x}} \cdot e^{\mu_1 n - n/2} \le k$$

$$n\bar{x} \cdot \ln\left(\frac{1}{2\mu_1}\right) + \left(\mu_1 - \frac{1}{2}\right) n \le \ln k$$

$$n\bar{x} \cdot \ln\left(\frac{1}{2\mu_1}\right) \le \ln k - \left(\mu_1 - \frac{1}{2}\right) n$$

$$\left[\mu_1 > \frac{1}{2} \Rightarrow \frac{1}{2\mu_1} < 1 \Rightarrow \ln \frac{1}{2\mu_1} < 0\right]$$

$$n\bar{x} \ge \frac{\ln k - \left(\mu_1 - \frac{1}{2}\right) n}{\ln\left(\frac{1}{2\mu_1}\right)}$$

(*) $C = \left\{\bar{x} \ge \frac{\ln k - \left(\mu_1 - \frac{1}{2}\right) n}{\ln\left(\frac{1}{2\mu_1}\right) \cdot n}\right\}$ is best critical region for H_0 vs $\hat{H}_1$

So, $\bar{x}$ is enough to define best critical regions for H_0 vs $\hat{H}_1: \mu = \mu_1$, and uniformly best crit. region for H_0 vs H_1.

b) All uniformly best critical regions can be defined of the form $\tilde{C} = \{\bar{x} \geq c\}$.

Indeed, for $\forall c$, and any μ_1, we can choose $k \geq 0$ such that

$$\boxed{x \in \tilde{C} = \{\bar{x} \geq c\} \text{ if and only if } \frac{L(0.5; x)}{L(\mu_1; x)} \leq k}$$

/this is the definition of ~~the~~ $\tilde{C}$ being uniformly best/

Indeed, $\frac{L(0.5; x)}{L(\mu_1; x)} \leq k$ when

$$\bar{x} \geq \frac{\ln k - (\mu_1 - 0.5)n}{\ln\left(\frac{1}{2\mu_1}\right)n} \quad (\text{by a}))$$

or when

$\bar{x} \geq c$, if we choose

$$k = \exp\left(\ln\left(\frac{1}{2\mu_1}\right)nc + n\left(\mu_1 - \frac{1}{2}\right)\right)$$

This k is positive since exponent is always ≥ 0.

c) $\alpha = 0.068$ Hint: $10\bar{X} \sim Poi(5)$ given H_0

$n = 10$ ~~$10\bar{X} \sim Poi(10 \cdot \frac{1}{2})$~~

Goal is find c, such that
$P(\bar{X} \geq c \mid H_0) = 0.068$

$$P(10\bar{X} \geq 10c \mid H_0) = P(\underbrace{Z}_{\sim Poi(5)} \geq 10c) = 0.068$$

or $(P(Z < 10c) = 0.932)$

From table III (Appendix)
it gives

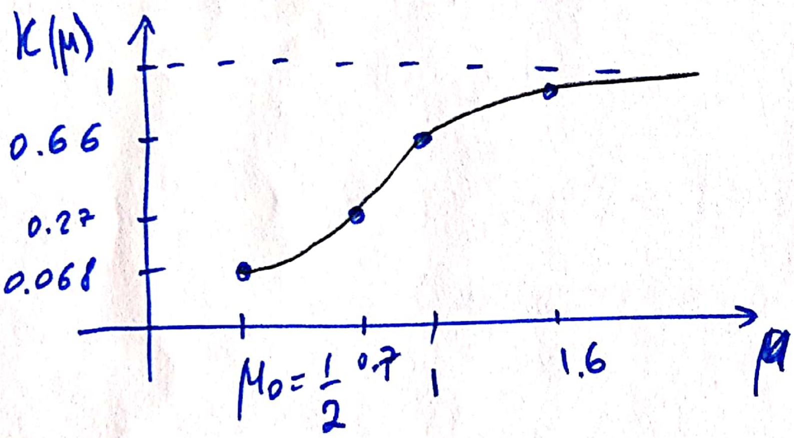
$10c = 8$ page 491, column 5.0

$\boxed{c = 0.8}$

Critical region $\{\bar{X} \geq 0.8\} \Rightarrow$ reject H_0

(makes sense since $0.8 > 0.5 = \mu_0$)

d) Power function



$K(\mu)$ = probability to reject H_0 when parameter is μ

$$K(\mu) = P(\bar{X} \geq 0.8 \mid \mu) = P(\underset{\sim Poi(10\mu)}{Z} \geq 8)$$

E.g. $\mu = 1$ $\quad K(\mu) = P(\underset{\sim Poi(10)}{Z} \geq 8) = 1 - 0.333 \sim 0.666$

$\mu = 0.7$ $\quad K(\mu) = P(\underset{\sim Poi(7)}{Z} \geq 8) = 1 - 0.729 \sim 0.27$

$\mu = 1.6$ $\quad K(\mu) = P(\underset{\sim Poi(16)}{Z} \geq 8) = 1 - 0.022 \approx 0.98$

④ p-value is tail probability, so it can take values only in $[0,1]$

We reject H_0 if and only if p-value $< \alpha$ = significance level = probability to reject H_0 if it is true

$P(\text{reject } H_0) \neq \alpha$, so just first sentence does not define α, so it is not enough to make a conclusion.

$$0.5 = P(\text{reject } H_0) = P(\text{reject } H_0 \mid H_0) \cdot P(H_0) + P(\text{reject } H_0 \mid H_1) \cdot P(H_1) = \alpha \cdot P(H_0) + (1-\beta) \cdot P(H_1) =$$

$$= \alpha \cdot (1-0.1) + (1-\beta) \cdot 0.1 =$$

$$= 0.9\alpha + 0.1(1-\beta) \le 0.9\alpha + 0.1$$

since $0 < \beta < 1$

$$0.5 \le 0.9\alpha + 0.1$$

$$5 \le 9\alpha + 1$$

$$4 \le 9\alpha$$

$$\frac{4}{9} \le \alpha$$

$\alpha \geq \frac{4}{9} > \frac{3}{10} = 0.3 = $ p-value, so

H_0 should be <u>rejected</u>.